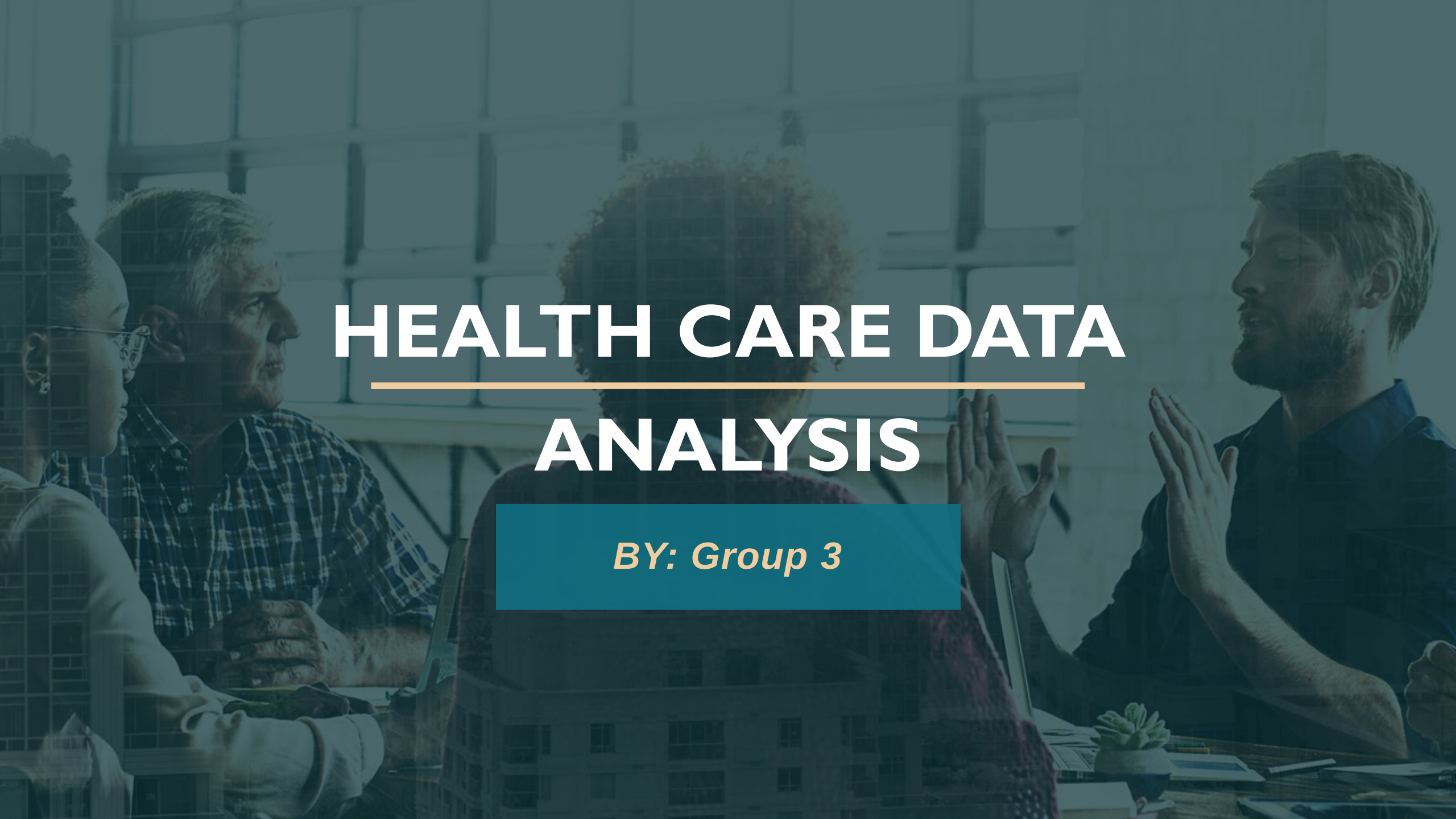


A group of four people (three men and one woman) are seated around a table in a meeting. The image is overlaid with a semi-transparent teal grid pattern. The title 'HEALTH CARE DATA ANALYSIS' is centered in white, with a thin orange line under 'HEALTH CARE DATA'. Below it, 'BY: Group 3' is written in orange on a teal rectangular background.

HEALTH CARE DATA ANALYSIS

BY: Group 3



PROJECT SUMMARY

To analyze healthcare data across various metrics including patient visits, diagnoses, treatments, test results, and doctor workload and to use the master tools like Excel , Power BI , Tableau and SQL. With the scope of comparative analysis across cities, test types, visit types, and insurance providers,

KEY PERFORMANCE INDICATORS

- ❑ AVG Patient Age: 49
- ❑ Total Doctors: 1,000
- ❑ Total Patient: 10,000
- ❑ Total Visit: 10,000
- ❑ Doctor Workload: 10 patient
- ❑ Most Common Visit Type: Specialist Consultation (25.59%)
- ❑ Top Diagnosis: Migraine (20.39%)
- ❑ Year with most visits: 2024 (4,927 visits)
- ❑ AVG Treatment cost: \$518.87-\$529.27

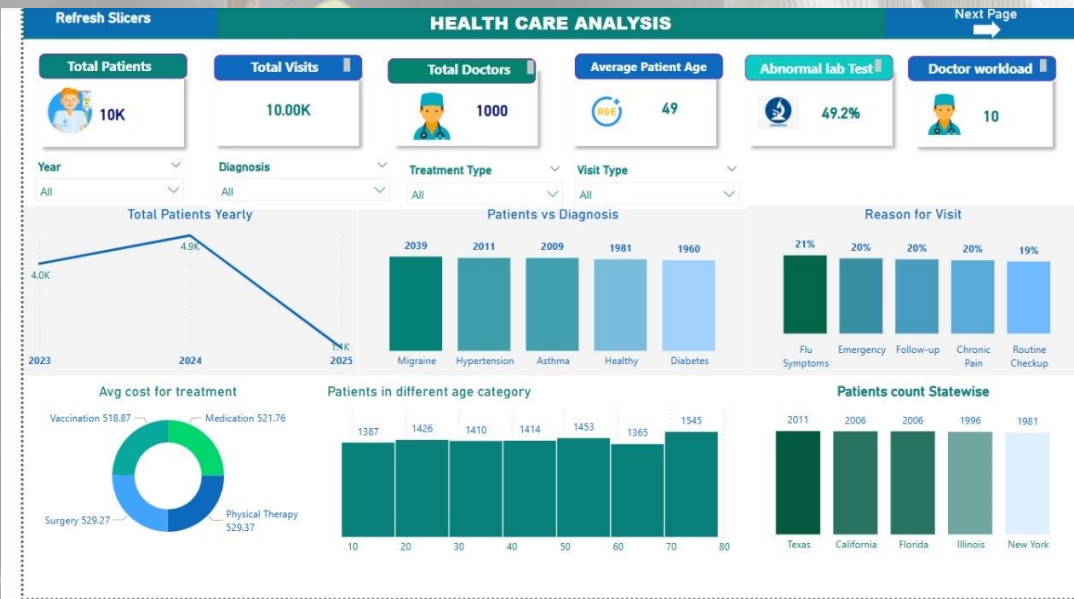
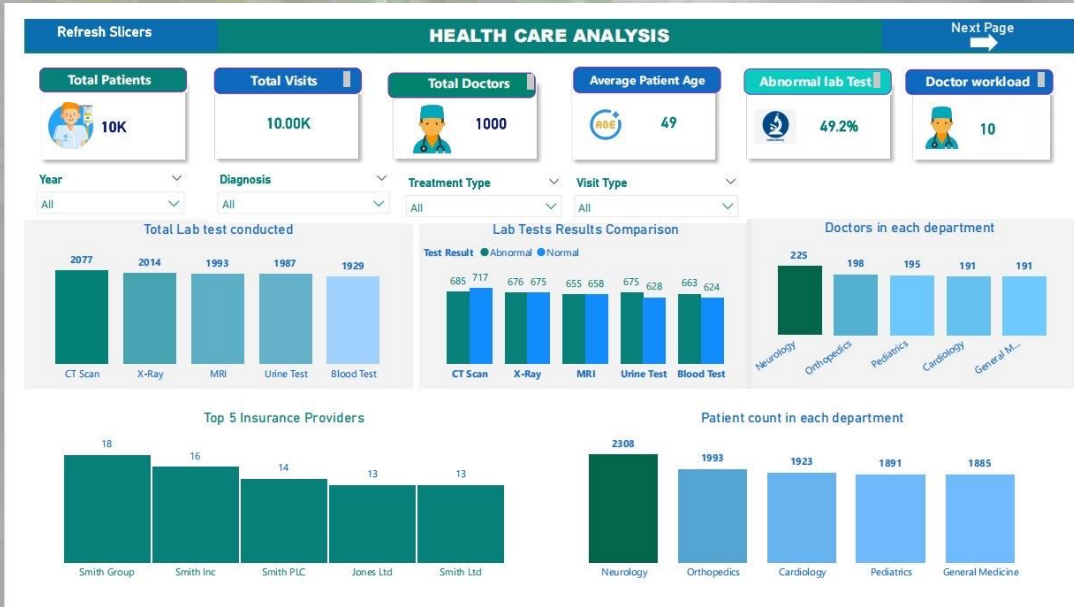
Excel Dashboard



➔ Insights from Dashboard ←

- Distribution of visit types and diagnoses
- High follow-up rate and abnormal test outcomes
- Lab results indicate potential quality control focus

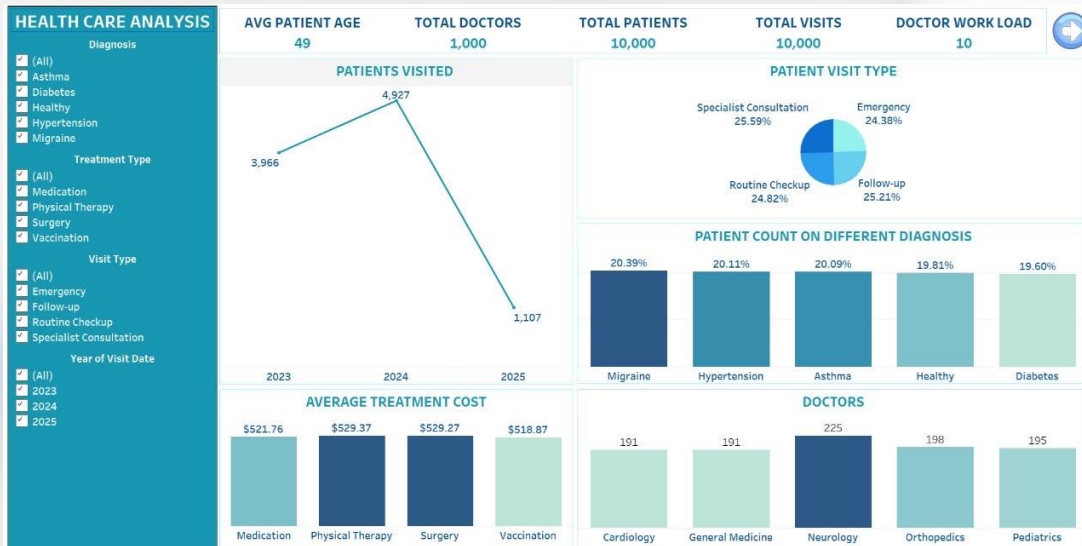
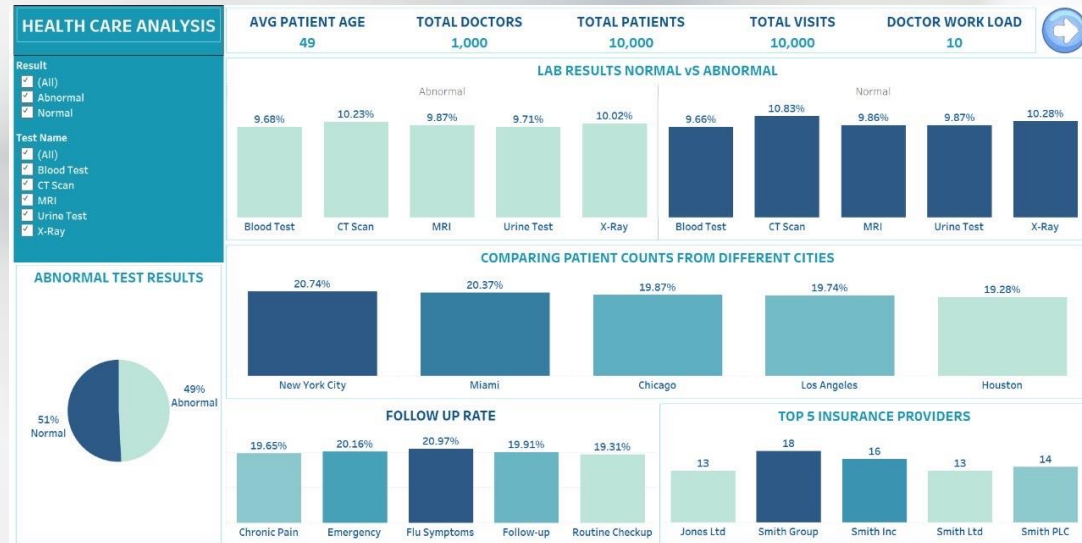
Tableau Dashboard



Insights from Dashboard

- ❑ Abnormal lab test rate: 49%
- ❑ Department-wise doctor distribution
- ❑ Cost comparison of top insurance providers
- ❑ Regional patient distribution analysis

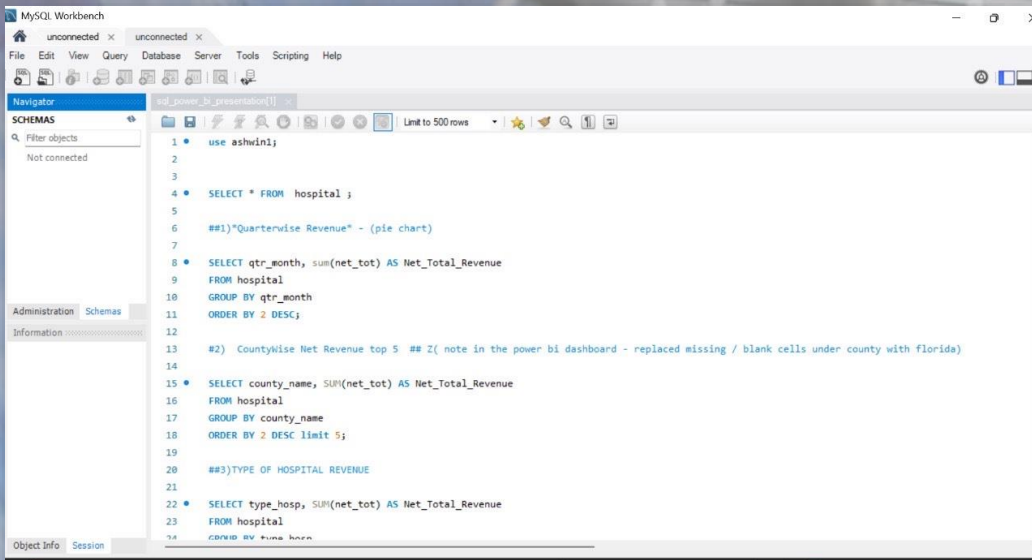
Power BI Dashboard



Insights from Dashboard

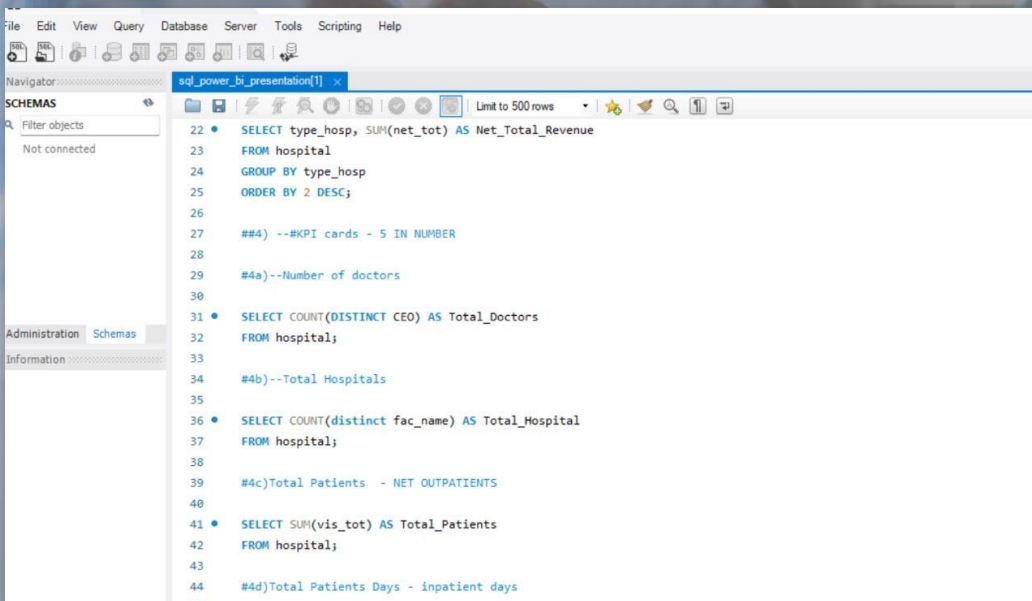
- ❑ Top cities with high patient volume: NYC & Miami
- ❑ Treatment cost variation across diagnosis types
- ❑ High patient count for chronic illnesses
- ❑ Deep dive into follow-up frequency

SQL Query



A screenshot of the MySQL Workbench interface. The 'Navigator' pane on the left shows the 'SCHEMAS' tab with a search filter. The main editor displays a SQL script with the following queries:

```
1 use ashwin1;
2
3
4 SELECT * FROM hospital ;
5
6 ##1) "Quarterwise Revenue" - (pie chart)
7
8 SELECT qtr_month, sum(net_tot) AS Net_Total_Revenue
9 FROM hospital
10 GROUP BY qtr_month
11 ORDER BY 2 DESC;
12
13 ##2) CountyWise Net Revenue top 5 ## Z( note in the power bi dashboard - replaced missing / blank cells under county with florida)
14
15 SELECT county_name, SUM(net_tot) AS Net_Total_Revenue
16 FROM hospital
17 GROUP BY county_name
18 ORDER BY 2 DESC limit 5;
19
20 ##3) TYPE OF HOSPITAL REVENUE
21
22 SELECT type_hosp, SUM(net_tot) AS Net_Total_Revenue
23 FROM hospital
24 ORDER BY 2 DESC;
```



A screenshot of the MySQL Workbench interface showing a different SQL script. The 'Navigator' pane on the left is the same. The main editor displays the following queries:

```
22 SELECT type_hosp, SUM(net_tot) AS Net_Total_Revenue
23 FROM hospital
24 GROUP BY type_hosp
25 ORDER BY 2 DESC;
26
27 ##4) --KPI cards - 5 IN NUMBER
28
29 #4a)--Number of doctors
30
31 SELECT COUNT(DISTINCT CEO) AS Total_Doctors
32 FROM hospital;
33
34 #4b)--Total Hospitals
35
36 SELECT COUNT(distinct fac_name) AS Total_Hospital
37 FROM hospital;
38
39 #4c)Total Patients - NET OUTPATIENTS
40
41 SELECT SUM(vis_tot) AS Total_Patients
42 FROM hospital;
43
44 #4d)Total Patients Days - inpatient days
45
```

Insights from Dashboard

- ☐ Patient age & workload
- ☐ Top diagnoses & treatment cost
- ☐ Follow-up rates & abnormal test ratios
- ☐ Lab test distribution

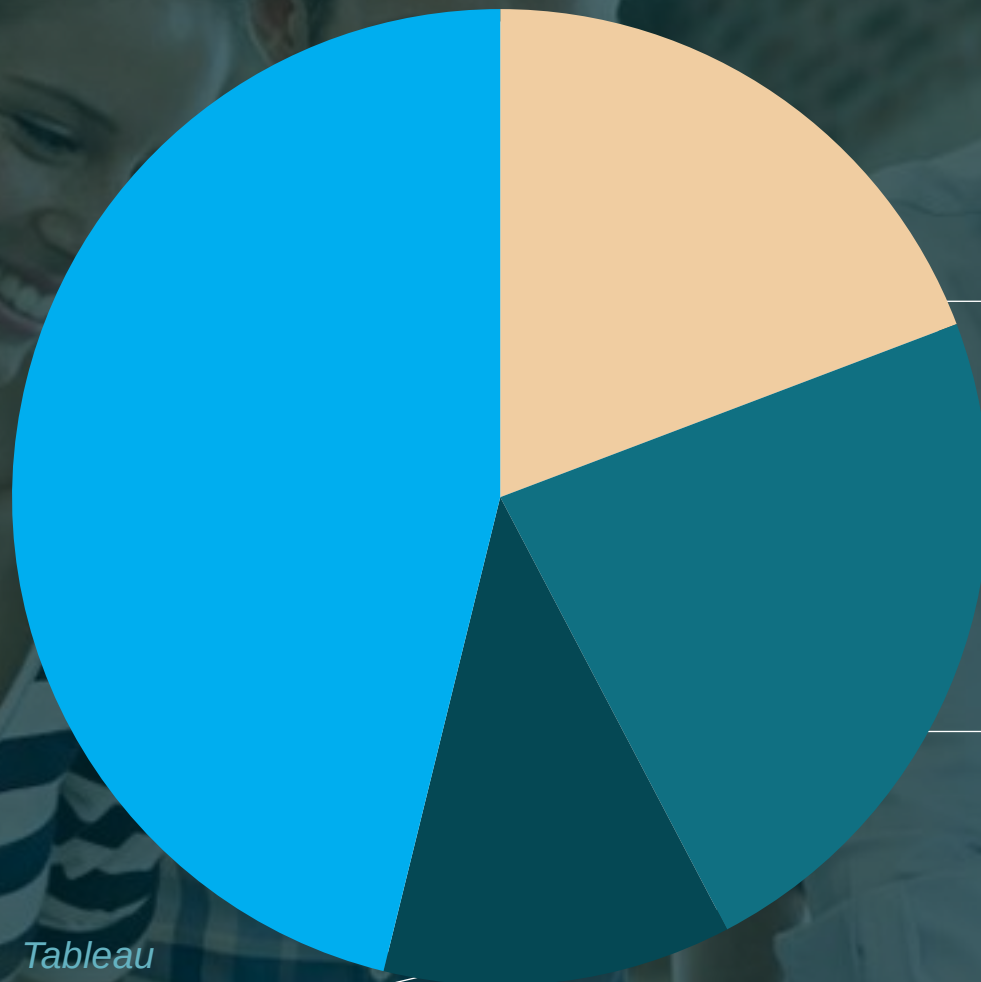
USE OF TOOLS

Excel
50-60%

Tableau
10-15%

Power BI
20-30%

SQL
40%



Key Insights & Recommendations

❑ **Demographics:**

Average patient age is 49, indicating a balanced population with both preventive and chronic care needs.

❑ **Workload Efficiency:**

~10 patients per doctor suggests manageable caseloads, supporting timely care and provider sustainability.

❑ **Follow-Up Visits (25%):**

A solid portion of visits are follow-ups, reflecting good continuity of care but also ongoing patient management need.

❑ **Abnormal Test Results (49%):**

High rate of abnormal results may point to elevated health risks in the population, requiring closer clinical oversight.

❑ **Top Diagnoses:**

Migraine, hypertension, and asthma are the most common—highlighting a need for focused chronic care pathways.

❑ **Recommendation:**

Prioritize resource allocation in NYC and Miami due to higher demand or clinical burden.

THE TEAM

- ❑ **Gayathry Prasad**
- ❑ **Nikhat Jarin**
- ❑ **Akul Srujanasheela**

Group 3

- ❑ **Komal Doolani**
- ❑ **Surabhi Basant Sahoo**

THANK YOU!

